Standford CS231n: Optimization

Score function: $f(x, W) \rightarrow$ Image Pixels $\rightarrow$ multiply by weights $\rightarrow$ scores

[1.2, 4.7, -0.8, 2.1] $\rightarrow$ The largest score is what the model predicts

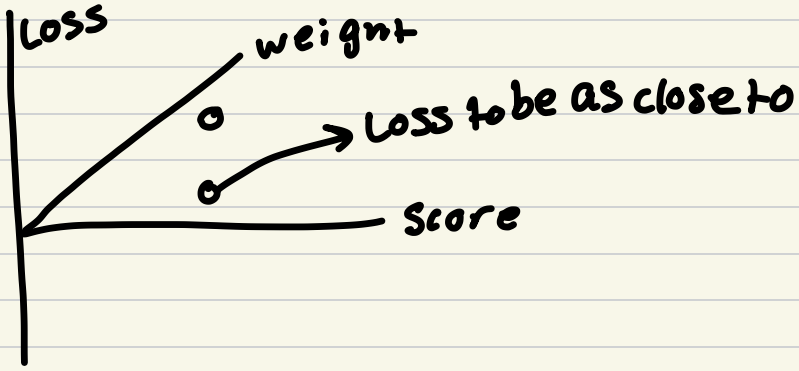
Loss function how bad are model weights

- Small = Good
- Large = Bad

Optimization

- Find value of weight that makes loss smallest

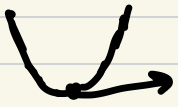
Loss = Landscape



$$L_i = \sum_{j \neq y_i} \max(0, w_j^T x_i - w_{y_i}^T x_i + 1)$$

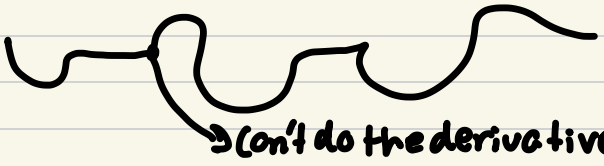
How close is the wrong class score with correct class score

Convex Functions



$\rightarrow$ Can find minimum where score = predicted

Neural Neural Network Non-Convex



$\rightarrow$ Can't do the derivative so use subgradient

Random Search

- Calculate random weights and keep whichever weights are best

Iterative Refinement: Start Random but keep readjusting weights

